

Rivera Wijaya

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EDUCATION

University of Toronto

BASc. in Electrical and Computer Engineering with PEY Co-op

Sept 2024 – Apr 2029

Toronto, ON

- **GPA: 3.92/4.00** Dean's Honours List for all semesters

EXPERIENCE

Firmware Engineer

University of Toronto Aerospace Team

May 2026 – Present

Toronto, ON

- Developed bootloaders for both single-bank and dual-bank MCUs using MCUBoot. Used external flash with image swapping for the single-bank MCU and direct XIP for the dual-bank MCU.
- Implemented a UART-to-I2C bridge on the Arty A7-100T; developed SystemVerilog testbenches in ModelSim to verify command decoding and I2C sequencing, with Tcl/.do scripts for automated compilation, simulation, and waveform debugging.

Power Electronics Engineer

Robotics for Space Exploration

Sept 2025 – Present

Toronto, ON

- Designing a configurable 4-switch buck-boost DC/DC converter platform for rover power distribution, supporting 5 V/8 A, 12 V/5 A, 24 V/5 A, and 55 V/0.55 A outputs from a 6S battery system on KiCAD.
- Validated power-stage components (MOSFET stress, gate charge, inductor sizing, feedback, efficiency, thermal).

Firmware Engineer

University of Toronto Robotics Association

September 2025 – January 2026

Toronto, ON

- Integrated IMU, ToF, and IR sensing over I2C and SPI with register-level bring-up and hardware sanity testing.
- Built Raspberry Pi motor-control and logging scripts; maintained GitHub-based documentation, issues, and pull requests.

PROJECTS

AXI-Based Real-Time Camera Event Detector

- Designed a custom AXI4-Stream hardware accelerator on Arty A7-100T for real-time camera/vision event detection (motion, edges, object centroid), with AXI4-Lite registers for threshold, mode, and status configuration.
- Authored streaming RTL for pixel framing, buffering, event encoding, and valid/ready backpressure; integrated into a Vivado block design with MicroBlaze control and UART reporting.
- Built a UVM verification environment with randomized frame traffic, AXI4-Stream backpressure, scoreboard-based checking, functional coverage, and SVA; used Vivado ILA for on-board waveform debug and cycle-level latency/throughput measurements.

TinyTPU 4x4 Systolic Matrix-Multiply Accelerator for Tiny Tapeout | [🔗](#)

- Designed a 4x4 output-stationary systolic array with 16 signed INT6 MAC processing elements, propagating operands across registered PEs while accumulating INT14 partial sums.
- Implemented persistent accumulation across K-tiles and serialized operand/result transfers, allowing matrices larger than 4x4 to reuse the same systolic array without external partial-sum storage.
- Built randomized cocotb regression against a Python golden model with SystemVerilog assertions; measured first-result latency, sustained throughput, and MACs/cycle to compare the baseline and persistent-accumulation architectures.
- Ran the RTL-to-GDSII ASIC flow with LibreLane targeting the SKY130A PDK, including synthesis, placement, routing, and GDS generation for Tiny Tapeout implementation.

Digital Theremin: Sensor-to-VGA + Audio RTL System | [🔗](#)

- Built an FPGA RTL system on the DE1-SoC that interfaced with an ultrasonic distance sensor, implemented threshold detection and FSM/datapath control, and generated VGA output through custom HS/VS timing and RGB pixel logic.
- Verified with self-checking testbenches, Quartus timing analysis, and board bring-up.

Real-Time 4-FSK Audio Modem | [🔗](#)

- Developed a real-time 4-FSK modem on the Cyclone V FPGA (De1-SoC) and Nios V soft core with a Goertzel-based DSP demodulator for tone detection and symbol decoding.
- Implemented packet synchronization and error-detection logic with preamble detection and parity checking; measured BER/SER and visualized receiver performance over VGA.

SKILLS AND COURSES

Languages: Python, Tcl, C/C++, RISC-V Assembly, Verilog, SystemVerilog, VHDL

Tools & Platform: Vivado, Quartus, ModelSim, UVM, SVA, Zephyr, Linux, Altium, KiCad, LTspice, Git

Hardware Courses: Computer Organization, Hardware Design, Digital Systems, Signals and Systems, Digital Electronics

Software Courses: Programming Fundamentals, Computer Architecture, Operating System, Data Structures and Algorithms